

Fundamentals of Reaction Mechanism.

Fundamental means Basic Principles,

- ↳ cleavage of Bonds
- ↳ Intermediates
- ↳ Types of attacking agents
- ↳ Types of Reactions

Reaction Mechanism (the way in which bonds are broken or formed)

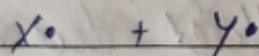
★ Bond Fission: (cleavage)

Homolytic fission
same

Heterolytic fission
different

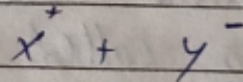
let an example of covalent bond $X-Y$

$X-Y$
after break down



Free radicals are formed

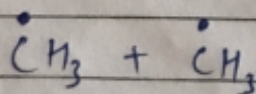
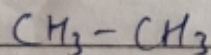
$X-Y$ → Let Y is more electronegative



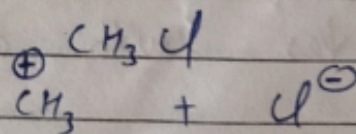
Cation

Anion form

example Ethane



Chloromethane



They are also called intermediates.

Intermediates Stability

- ↳ Free Radicals $\propto$ +I (inductive) groups
- ↳ Carbocations $\propto$ +M, +H, +I groups
Prefer first $\rightarrow$ mesomeric $>$ hyperconjugation $>$ inductive
- ↳ Carbanions $\propto$ -M, -I groups

* Inductive effect: An inductive effect is an induced or electronic effect due to the polarisation of σ bonds within a molecule or ion. This is typically due to an electronegativity difference between the atoms at either end of the bond.

$$\begin{array}{ccccccc} + & + & + & + & + & & - \\ \text{C} & - & \text{C} & - & \text{C} & \Rightarrow & \text{C} & \Rightarrow & \text{Cl} \end{array}$$

(Cl is more electronegative than carbon)

This induced effect only upto 3 to 4 carbon after this I effect is not so much.

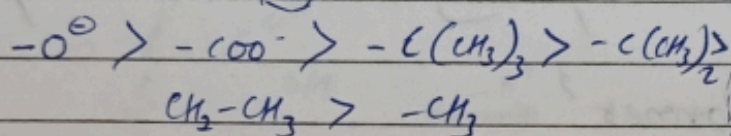
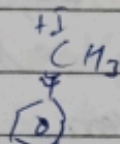
- ① it generates due to polarisation of bond.
- ② it also called transmission effect due to polarisation.
- ③ Polarisation is permanent so also called permanent effect.
- ④ only σ Bond involved not π Bond electrons.
- ⑤ Yes with distance.

Types of Inductive effect

+I effect

positive inductive effect occurs when the substituent donate electrons towards the chain

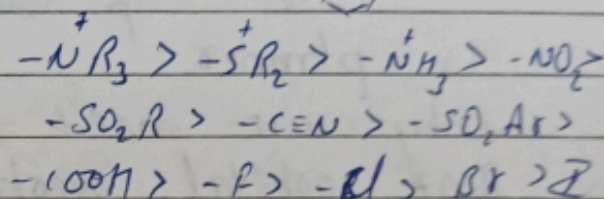
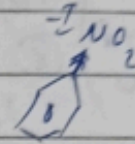
example



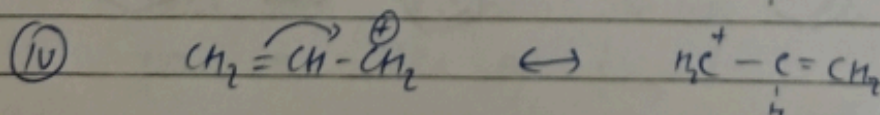
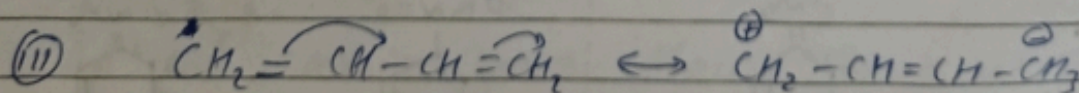
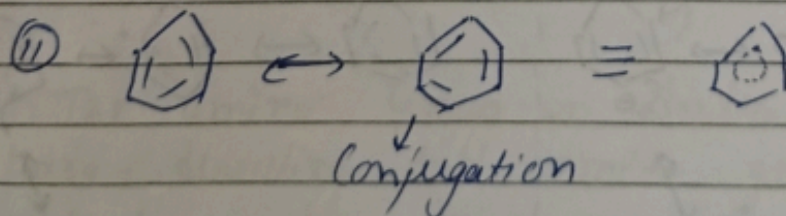
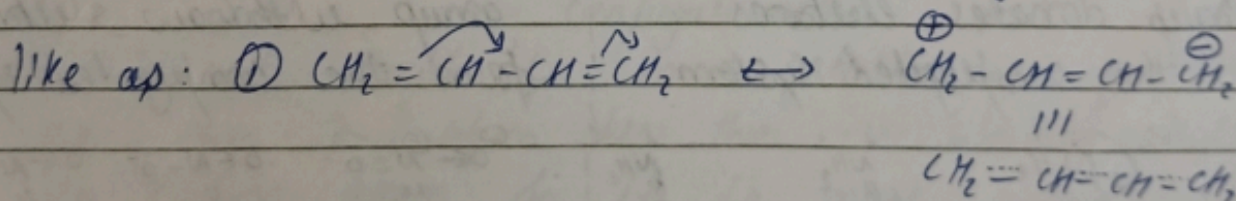
-I effect

negative inductive effect occurs when the substituent withdraws electron from the chain

example

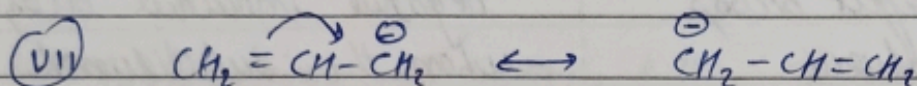
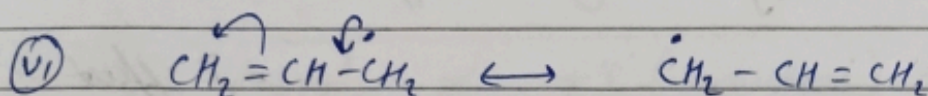
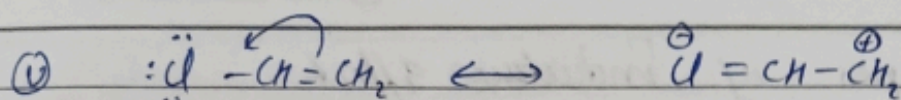


RESONANCE It is a way of describing delocalized electrons within a certain molecule or polyatomic ions where the bonding cannot be expressed by a single Lewis formula. It has two or more structures are possible difference only in the distribution of electrons.



NOTE $\rightarrow$ when free radical take part in conjugation it do not leave +ve charge or -ve charge.

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(NOTE - Important condition)
① Conjugation only possible when $-, =, -, =$ formate.
② All atoms lie in a same plane.
i.e sp^2 carbon hybridization. Bcoz sp^2 is planar.

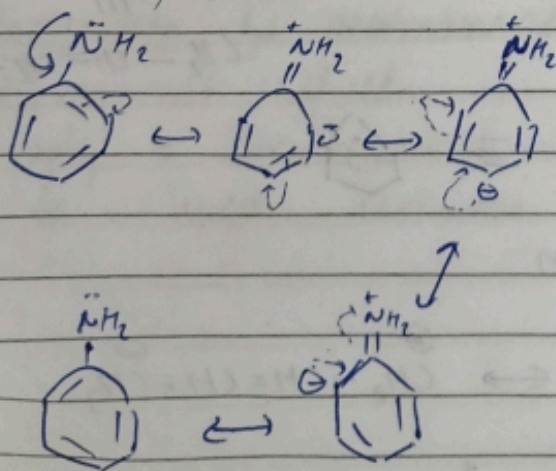
why we need planar?

Bcoz orbital overlapping become parallel to each other

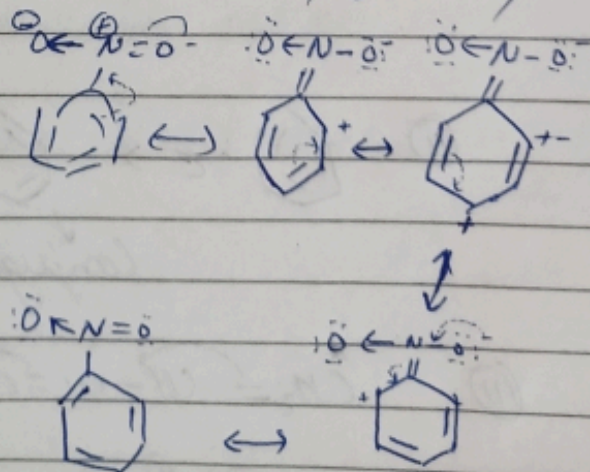
In sp^2 there is a vacant p-orbital, so e^- distribution is very easy due to vacant p-orbital

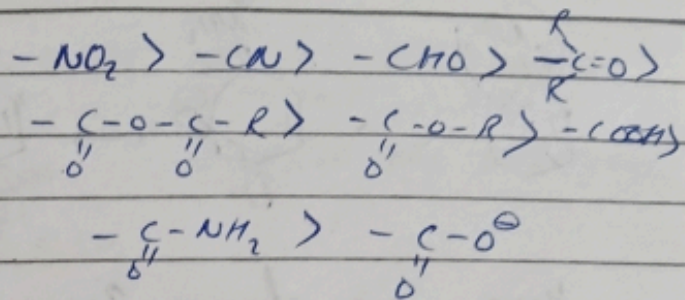
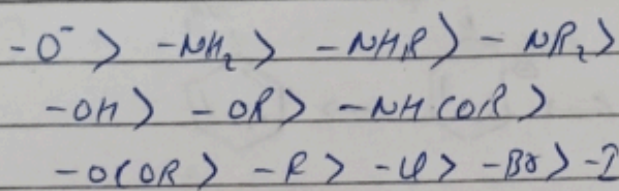
Resonance effect OR Mesomeric effect

positive +M effect
group donates electrons
to the conjugated system

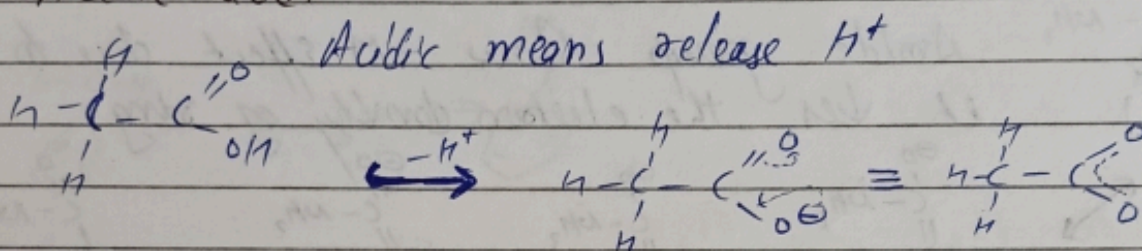


Negative -M effect
group withdraw electron
from the conjugated system





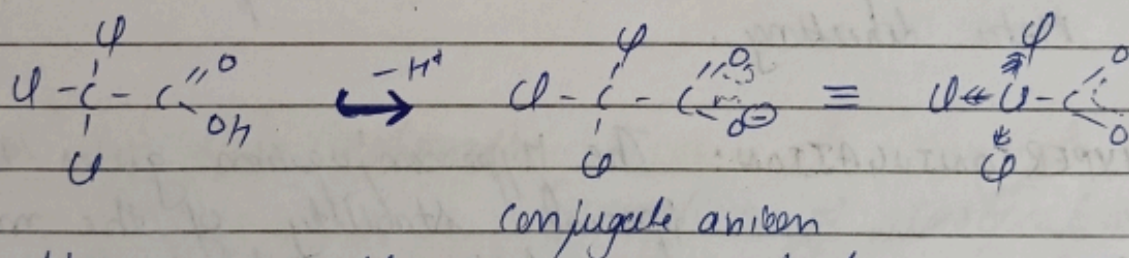
Question Explain the following.
Trichloroacetic acid is stronger Acid than Acetic acid.



For acidic character

conjugate anion

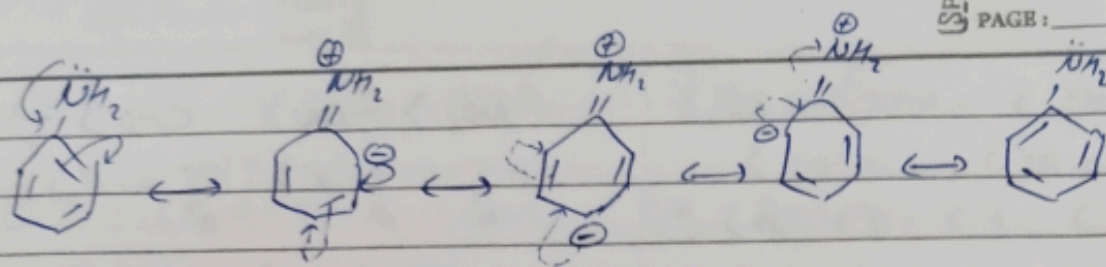
First we have remove acid H^+ then conjugate anion is formed i.e. overall the compound which have lesser -ve charge then the compound is more stable. or more acidic.



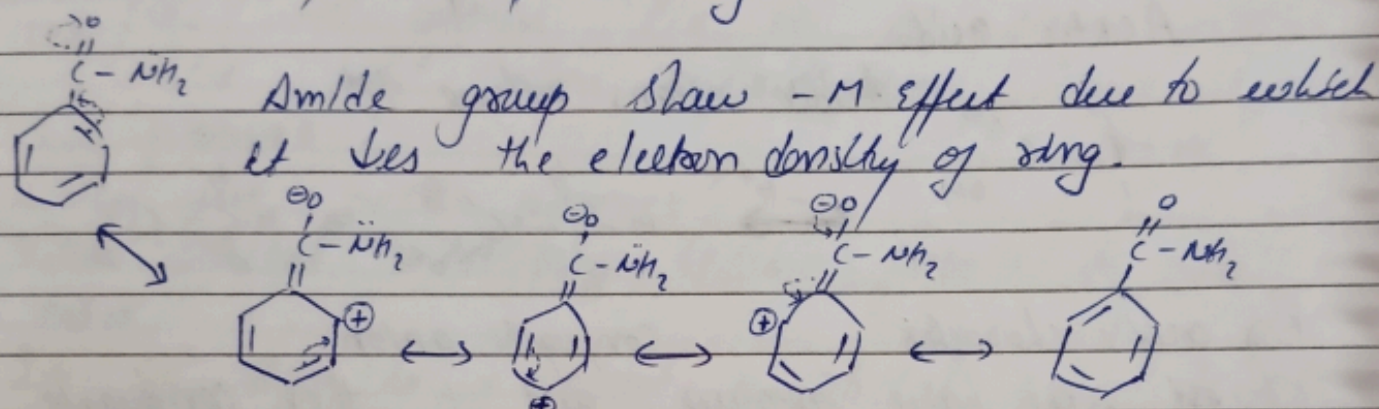
Trichloroacetic acid is more stable conjugate anion as the 3 Cl atom ves the -ve electron density on carbon atom. so it is more acidic in nature.

Question The amino group in Aniline is ortho and para-directing but amide group is meta directing.

As the amino group shows +M effect so it ves the electron density on ortho and para position. like as $\rightarrow$



Here -ve charge is at ortho position or para position so here if electrophile attacks it (E^+) attack O & p position i.e. amino is O & p-directing.



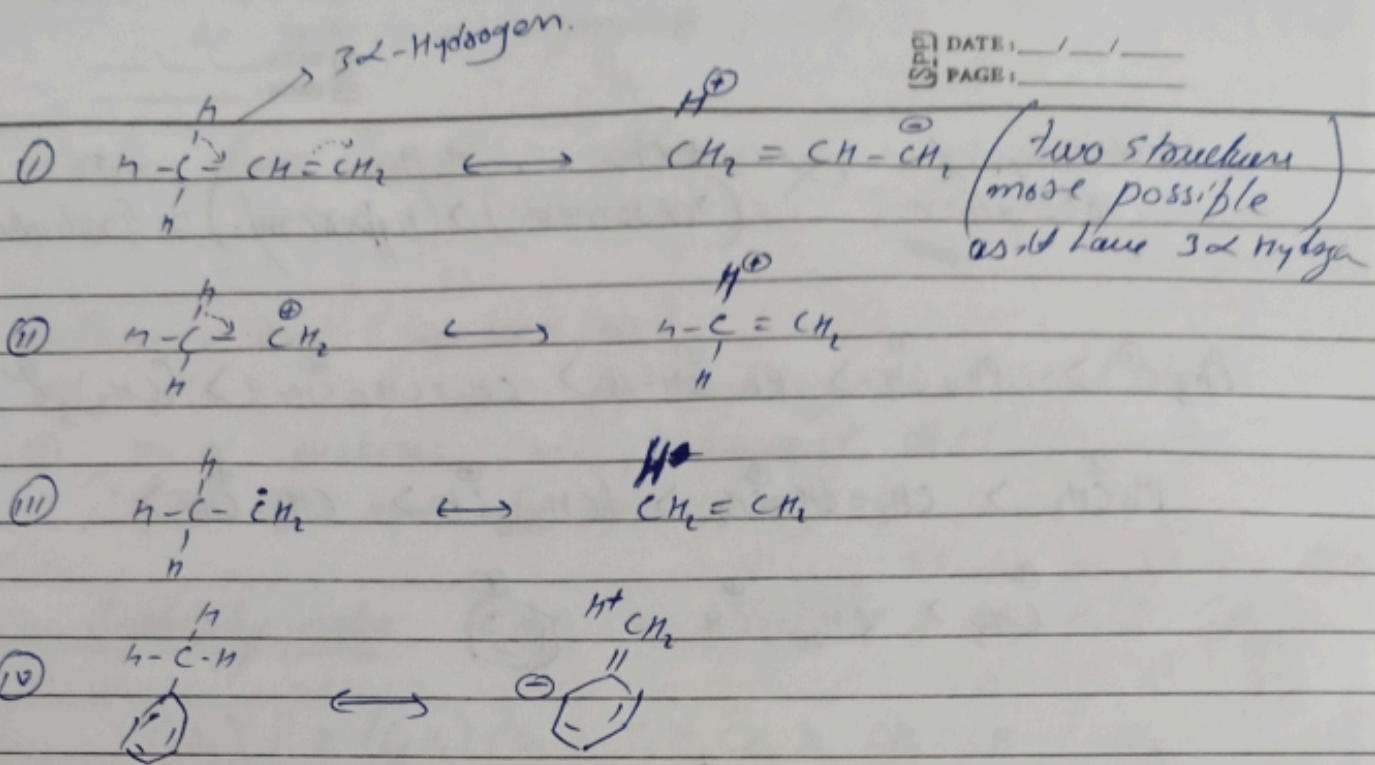
Now here we observe the electron density is at ortho and para position as compare to meta position so the electrophile attacks on meta position so it is meta directing.

HYPERCONJUGATION: The Hyperconjugation gives the special stability of the molecules, because one sigma electron delocalisation to form bonding or antibonding pi bond in conjugated carbon atom.

This Hyperconjugation is also called "no bond resonance"

(Delocalisation of ^{only} sigma electrons with p-orbital)

- Alkenes
- Alkynes
- Carbocation
- Free radical
- Benzene nucleus

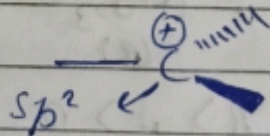


NOTE ① The methyl group show +I effect it is only because of Hyperconjugation.

② In Inductive effect the delocalisation is done only in sigma electron not double (π) or free electrons.

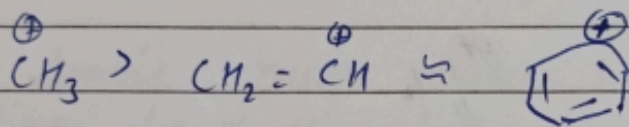
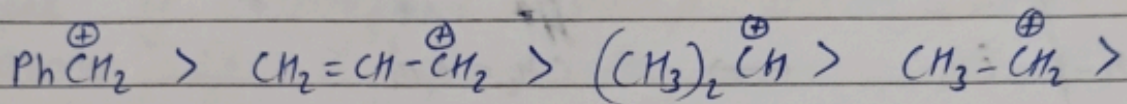
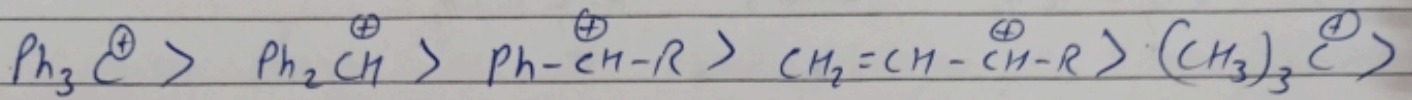
Stability of Carbocation

Formation of carbocation

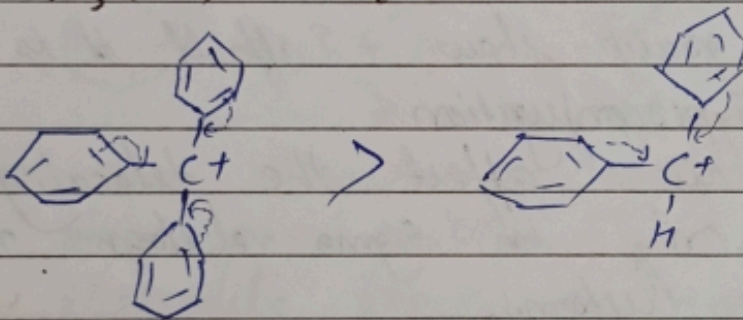


- i.e. only bond three σ Bonds (no lone pair)
- $3\sigma + 0\text{lp} = \text{sp}^2$ (hybridization) so
- Carbocation geometry/structure is Trigonal planar
- No. of electron in outermost shell = $6e^-$ (as 3 σ Bond. (1 σ include $2e^-$)

Stability of carbocation $\propto$ $\begin{matrix} +M & , & +H \\ \text{(mesomeric)} & & \text{(Hyperconjug.)} \end{matrix} > +I > \text{Inductive}$

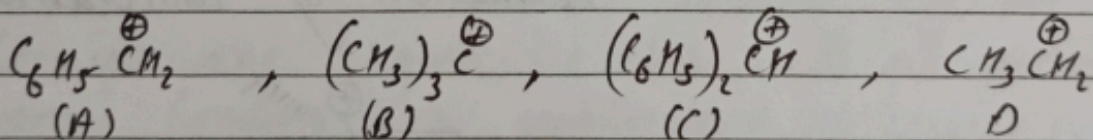


Example. $\text{Ph}_3\text{C}^+ > \text{Ph}_2\text{CH}^+$

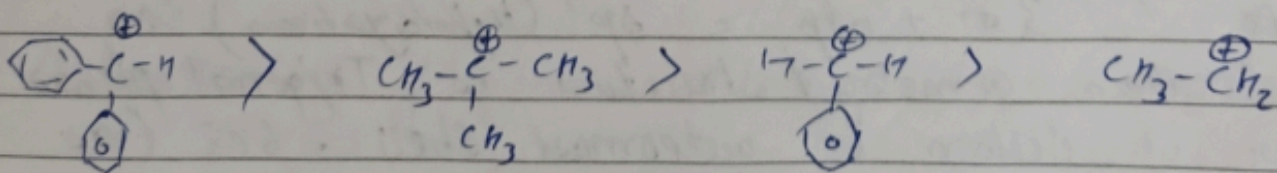


more stable

Ques. Arrange the order of their stability



Ans. Order $\text{C} > \text{B} > \text{A} > \text{D}$



NOTE - Carbocation or Free Radical they both are electron deficient as both have incomplete octet.

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Carbon Free Radical: Formation

① Three σ Bond (no lone pairs)

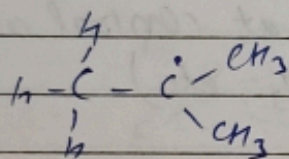
② $3\sigma + 0lp = sp^2$ (hybridization) Trigonal planar

③ no. of electrons in outermost shell is $7e^-$
(as 3σ and one free radical)

Stability order $(CH_3)_3\dot{C} > (CH_3)_2\dot{C}H > CH_3\dot{C}H_2 > \dot{C}H_3$

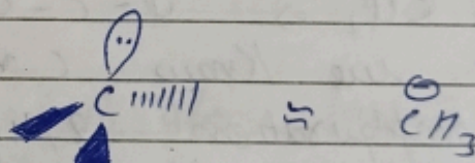
$(C_6H_5)_3\dot{C} > (C_6H_5)_2\dot{C}H > C_6H_5\dot{C}H_2 > CH_2=CH-\underset{\substack{| \\ CH_3}}{\dot{C}}-CH_3$

example



9 α H and more α H more is resonating structure more is the stability.

Carbanion: Formation



① Three σ Bond & 1 lone pair

② $3\sigma + 1lp = sp^3$ (hybridization) Pyramidal

③ no. of electrons in outermost shell is $8e^-$
(as 3σ and one lp)

Stability order $CH_3^- > (CH_3)-CH_2^- > (CH_3)_2-CH^- > (CH_3)_3C^-$

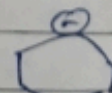
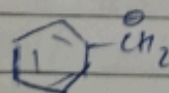
$CH \equiv C^- > CH_2 = \overset{\ominus}{C}H > CH_3-\overset{\ominus}{C}H_2$

%s = 50%

%s = 33.3%

%s = 25%

more is the %s character more is stability



In CH_3^+ resonance is possible while next no resonance so they predict on the basis of hybridization.

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NOTE $\Rightarrow$ overall if electron density $\uparrow$ towards the side of carbocation i.e stability $\uparrow$

if electron density $\uparrow$ towards the side of carboanion i.e stability $\uparrow$ as $-ve$ charge repels $-ve$ charges.

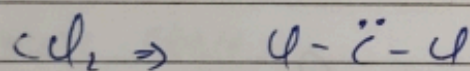
Type of Reagents:

(A) Electrophiles $\rightarrow$ electron loving species.

Positively charged species $\rightarrow \text{H}^+$, CH_3^+ , Br^+ , I^+ , NO_2^+

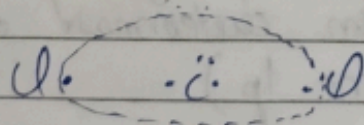
Neutral species $\rightarrow$ vacant orbital at central atom.
(BF_3 , SO_2 , BH_3 , etc)

Ques. why are Br^+ and Cl_2 electrophiles?



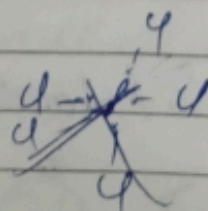
As we know C needs have its electronic configuration 2, 4 and it requires 4 electron to complete its octate.

But here, carbon have 6 e^- in octate



So it required 2 more electron to complete its octate i.e 2 electron deficiency i.e it is (Cl_2) is electrophiles

same reasoning. ~~its~~



Br^+ is a electron deficient compound so it acts as electrophile.

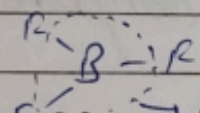
⑧ Nucleophiles: +ve charge loving species

→ Negative charged species = CN^- , OH^- , Br^- , I^-

→ Neutral species: NH_3 , H_2O etc.
i.e. lone pair.

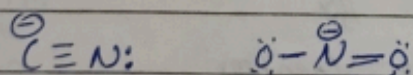
Ques: which is not nucleophile?
 H_2O BF_3 NH_3 OH^-

① water has lone pair so it is nucleophile

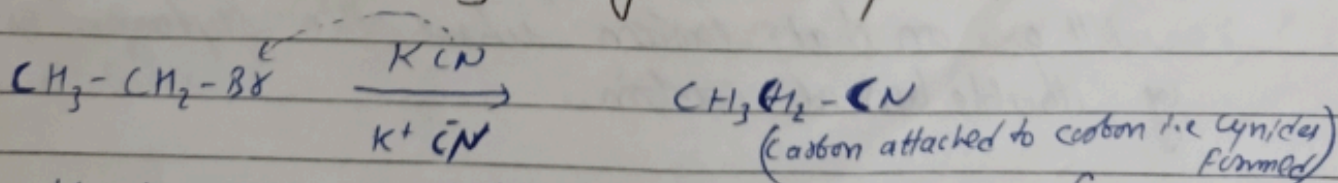
② BF_3  $B = 4, 3 = 2, 3$
 $6e^-$ But B also need $2e^-$ to complete its octet so it is electrophile.

③, ④ Both are nucleophile.

→ Ambident Nucleophile i.e. Both atoms have tendency to attack as a nucleophile.
example CN^- , NO_2^- etc.



Ques: Halalkanes on reacting with KCN give cyanides and with $AgCN$ give isocyanides.



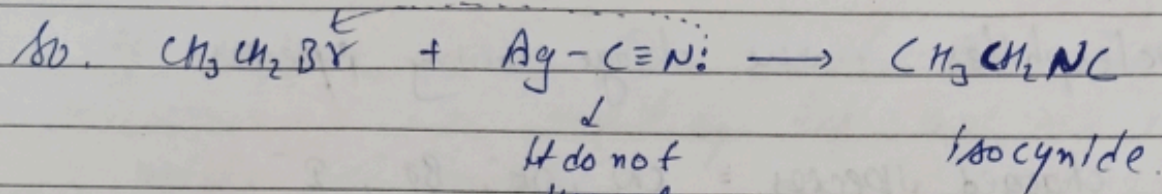
K is more metallic in nature than Ag (more metallic i.e. it is a ionic compound so it dissociate)
Good Write $K^+ + \bar{CN}$

2° carbocation more stable as compare to 1° carbocation.

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Meanwhile AgCN is a covalent compound

$\text{Ag}-\text{C}\equiv\text{N}:$ so it does not simply ionized



It does not
dissociate.
So N lone pair
attacks to Br.

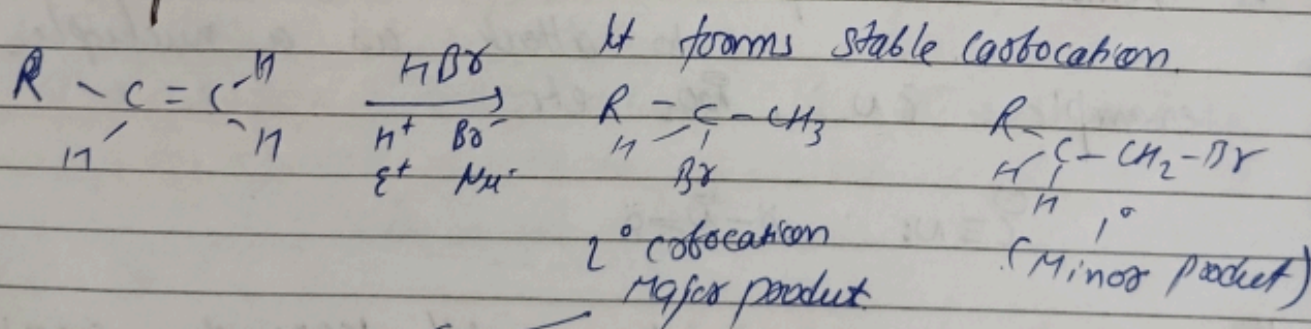
Isocyanide.

Types of Reactions:

(A) Addition Reaction: For addition we require unsaturation compound or carbonyl grp.
 $=, \equiv, >\text{C}=\text{O}$

- Electrophilic addition Reaction (add electrophile)
- Nucleophilic addition Rn^+ (" nucleophiles")
- Free radical addition Rn^+ (" of $\cdot\text{R}$)

Electrophilic addition Rn^+

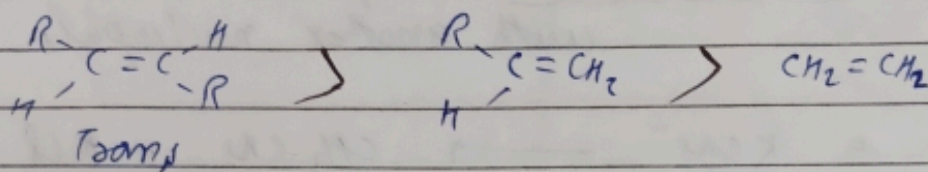
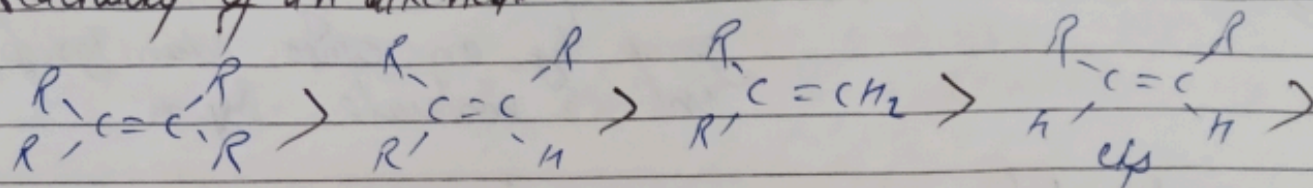


This is also called Markownikoff's product: (Br (Halogen)) will go on that carbon where the Hydrogen is less of double bonded carbon.

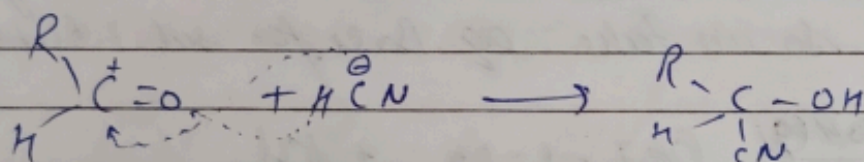
2° free radical is more stable than 1°

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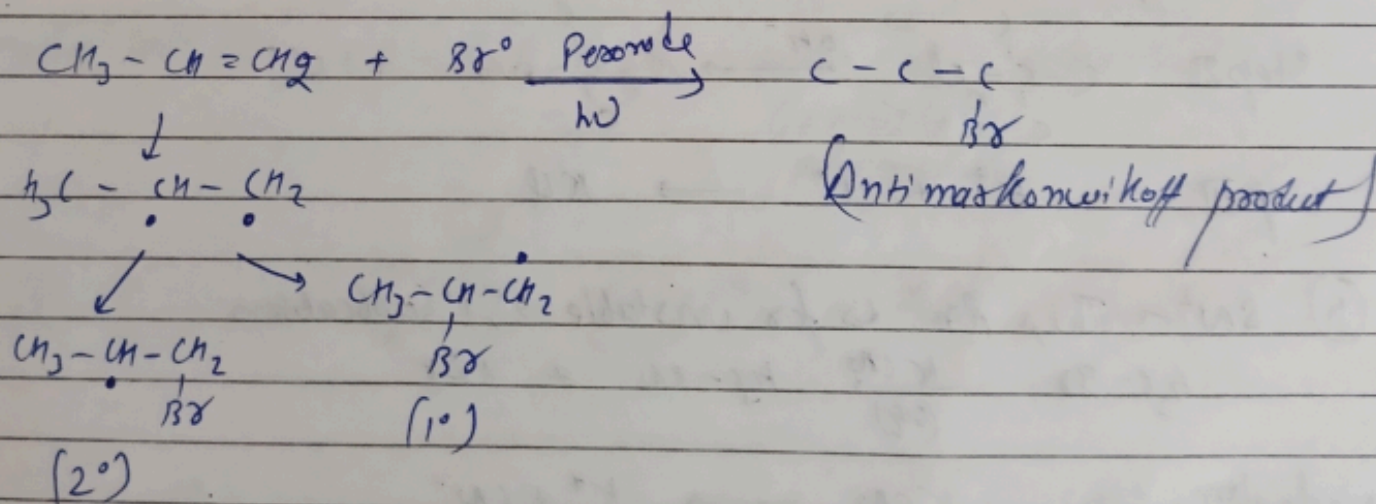
Reactivity of an alkene:



Nucleophilic Addition Reactions: Generally these Rn^+ show in carbonyl compounds.



Free Radical Addition Reactions:



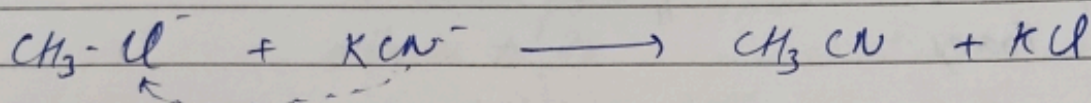
Question: Addition of HBr to alkene does not follow markownikoff's rule in presence of peroxide why?
Answer:

S_N1 always for 3° carbocation and S_N2 is always $2^\circ, 1^\circ$ carbocation (unstable)
(stable)

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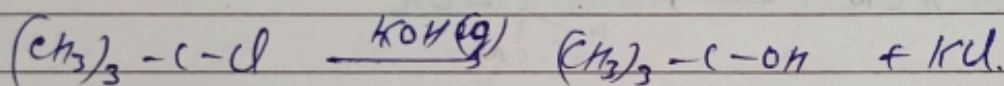
(B) Substitution Reaction: Replacement of one atom/group by an other atom/group in a molecule. By a

Nucleophilic Substitution R_n (Replacement one nucleophile with another nucleophile)



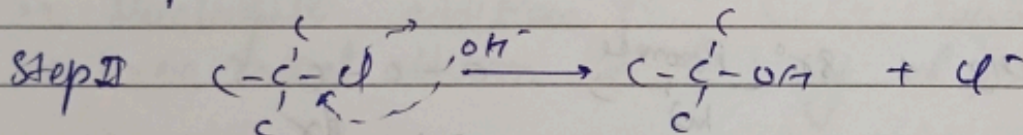
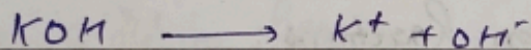
(1) S_N1 → This reaction is for stable 3° carbocation.

NOTE → Here we cannot take alc. Base bcoz it undergoes elimination R_n so we take aq. Base for substitution.

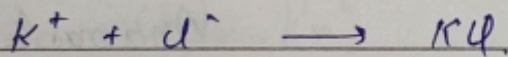


Mechanism

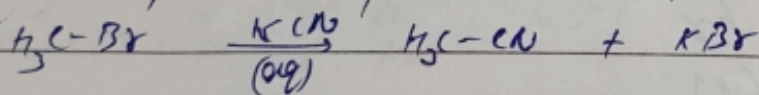
Step I



Step III

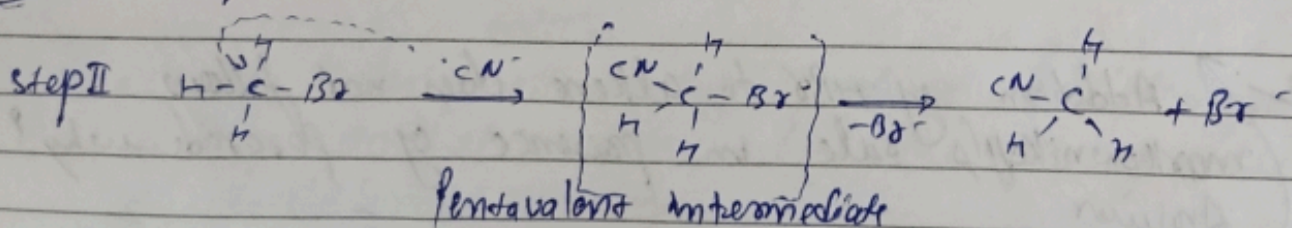
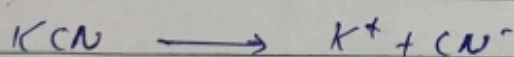


(ii) S_N2 → This R_n is for unstable $2^\circ, 1^\circ$ carbocation.

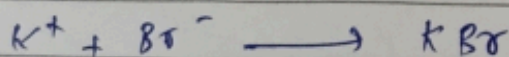


Mechanism

Step I



Step III

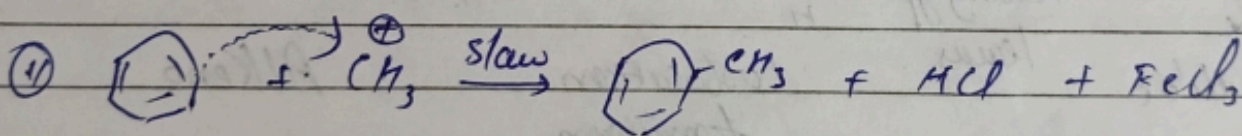
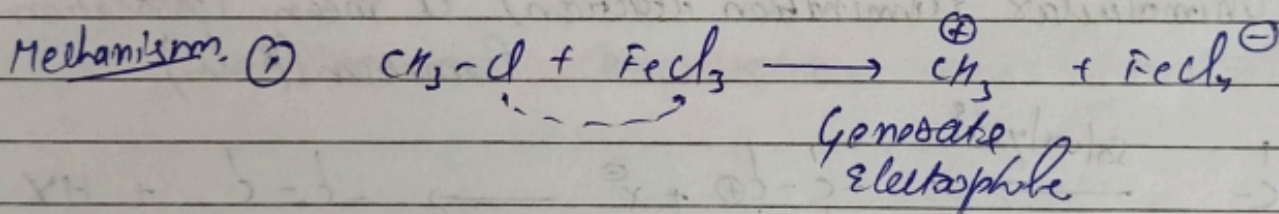
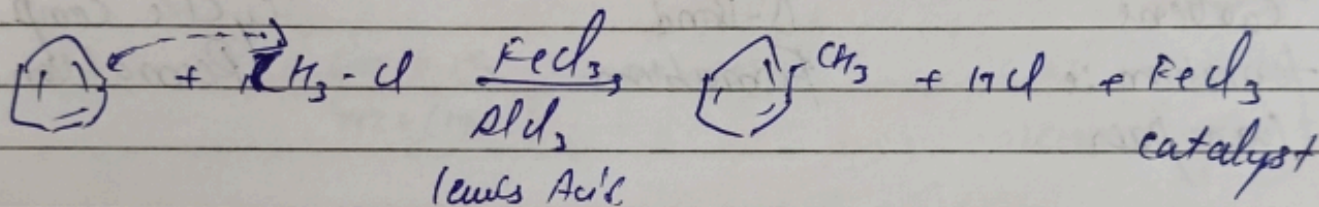


NOTE $\rightarrow$ Rate of $S_N1 >$ Rate of S_N2
 Bcoz 3° Carbocation is very stable as compare $1^\circ, 2^\circ$
 and in S_N2 slowly removal of nucleophile.

Stability order $3^\circ > 2^\circ > 1^\circ$ for S_N1 }
 $1^\circ > 2^\circ > 3^\circ$ for S_N2

Electrophilic Substitution Reaction: i.e. (e^+ attack's)

like as. Friedel-Crafts Reactions.



Another NOTE $\rightarrow$ S_N2 they are Inversions i.e. they are optically active.

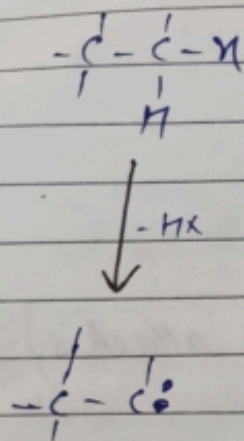
Dento compound $\xrightarrow{\text{change}}$ Levo compound
 Levo compound $\xrightarrow{\text{change}}$ Dento compound

Y There is no change i.e. Dento $\rightarrow$ Dento we call it retention.

Y Dento $\rightarrow$ Dento + Levo it is Racemic mixture.

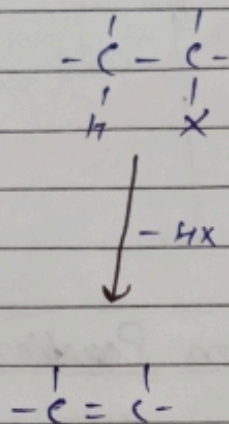
(C) Elimination Reaction: Removal of two atoms/groups

α -elimination



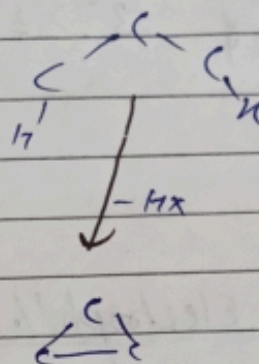
Carbene formation i.e. lone pair

β -elimination



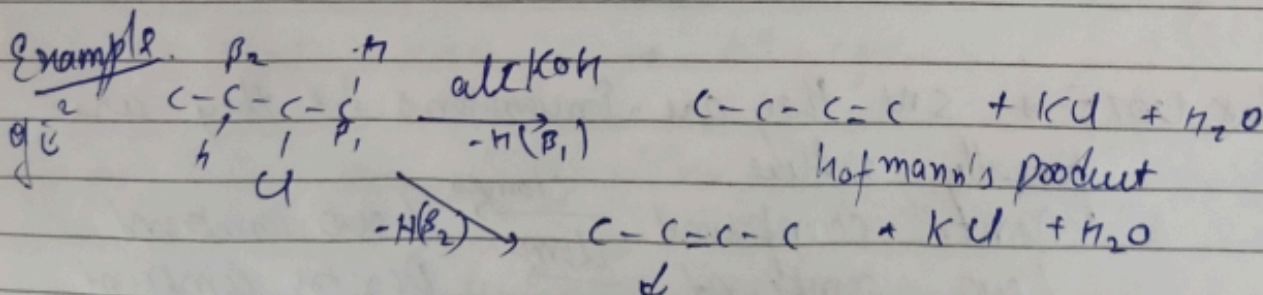
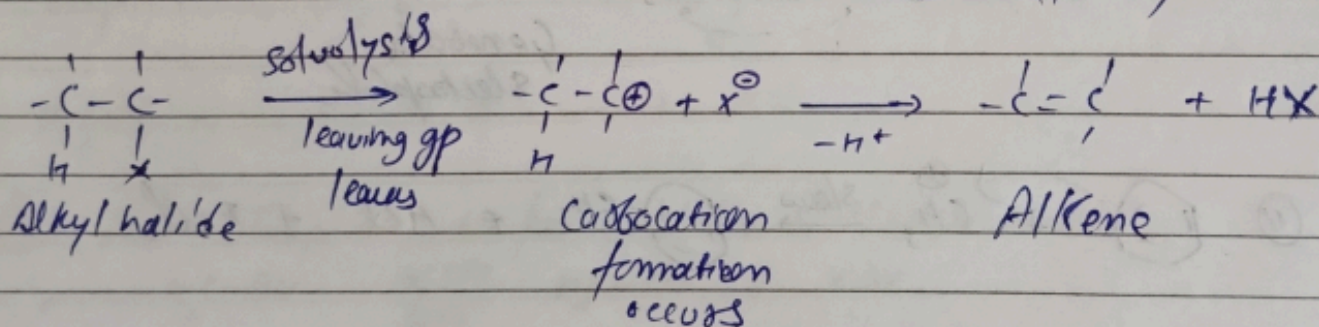
π -Bond Formation

γ -elimination



Cyclic comp. formation

E^1 (Unimolecular elimination reaction) E^1 mean carbocation formation (β -elim.)

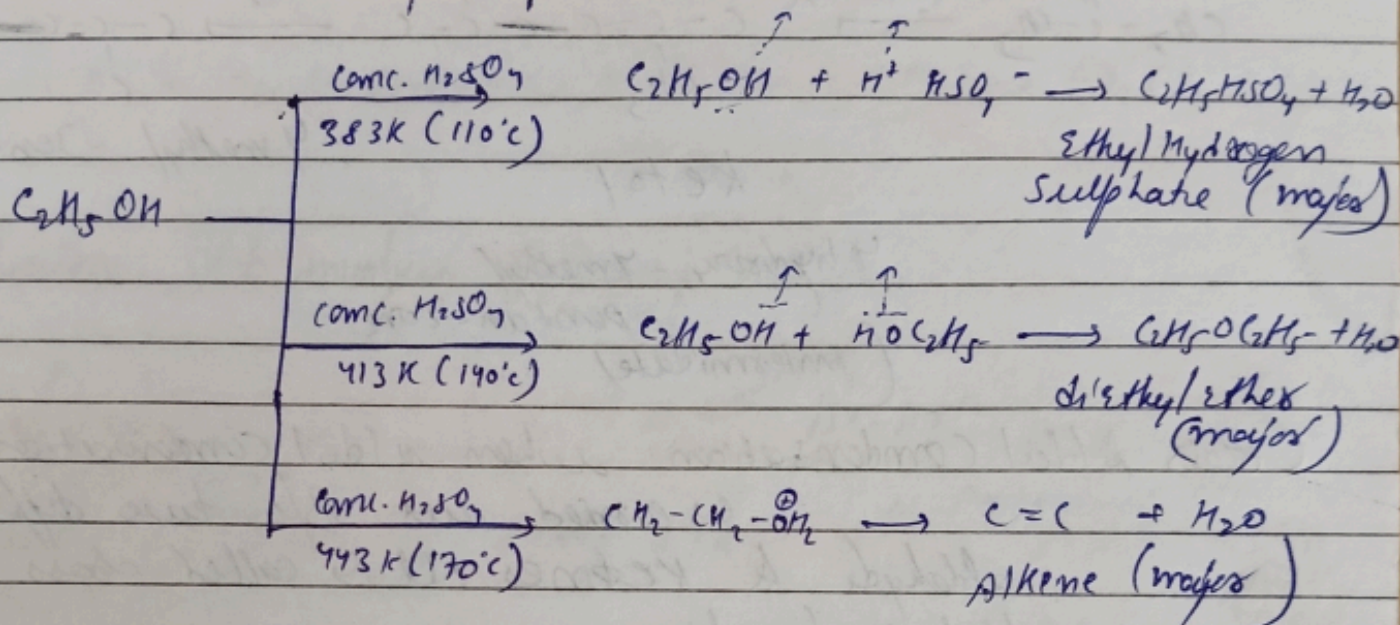


This is major product as two alkyl gp formed. (2 methyl)
Saytzeff product

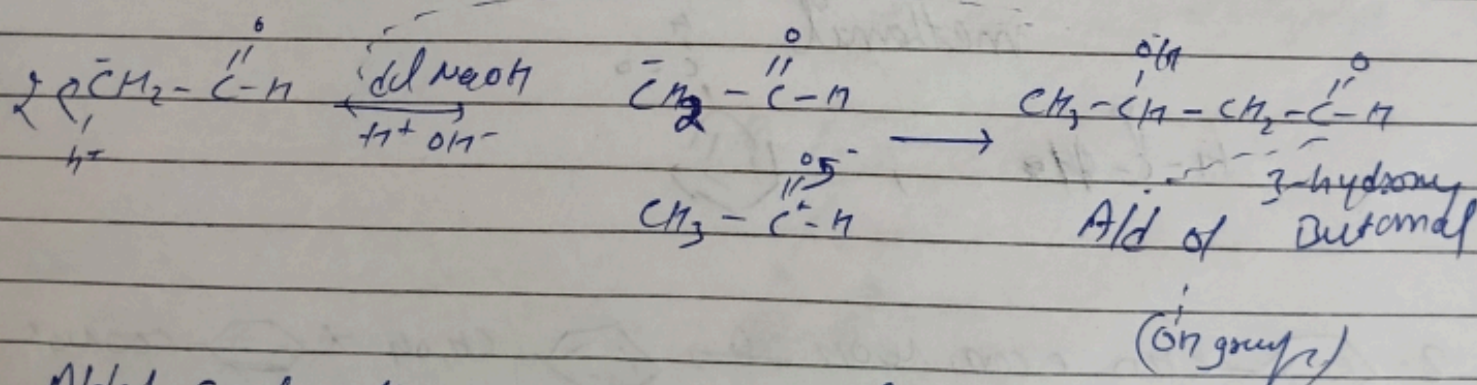
Ques write the product:

$\text{CH}_3\text{CH}_2\text{OH}$ heated with H_2SO_4 :

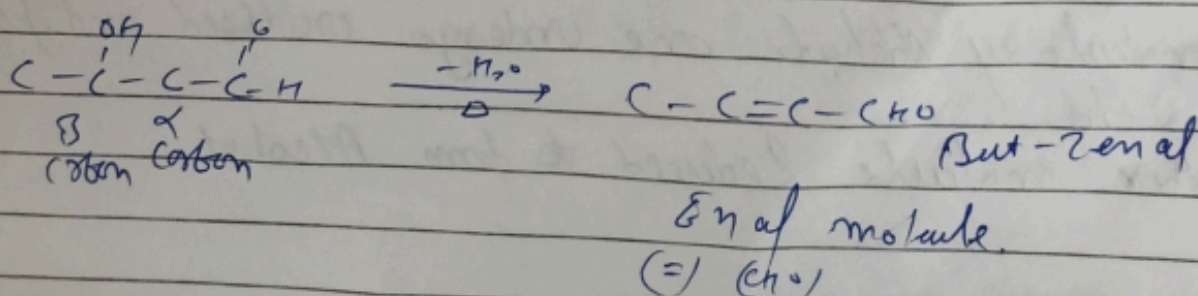
H_2SO_4 is a dehydrating agent. But this particular reaction $\text{C}_2\text{H}_5\text{OH}$ is temp dependent



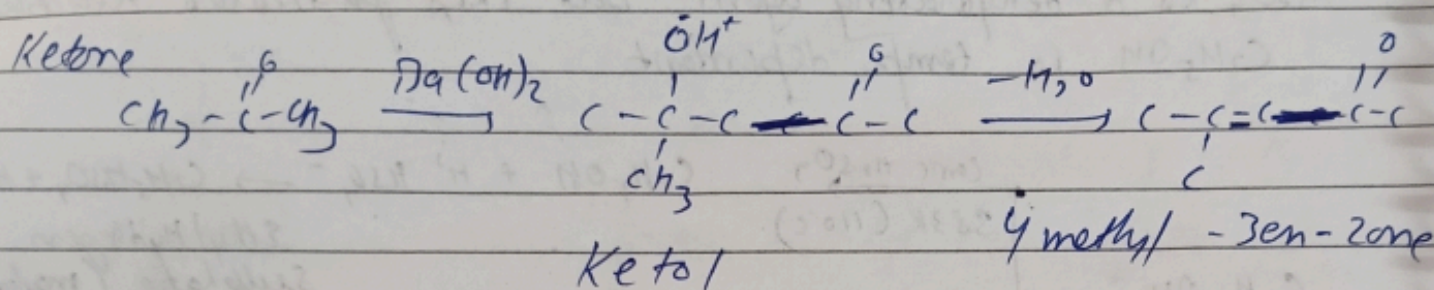
Aldol condensation: Condensation means (removal of H_2O) with Δ .



Aldol condensation is the removal of water from Aldol.



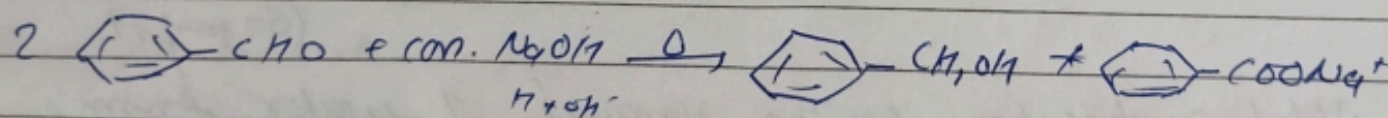
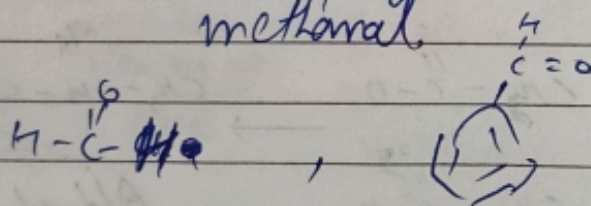
Why = occurs on α carbon:
Bcoz it is more stable on α -carbon. more
resonating structure.



4 hydroxy - 4 methyl
pentan - 2 one
(intermediate)

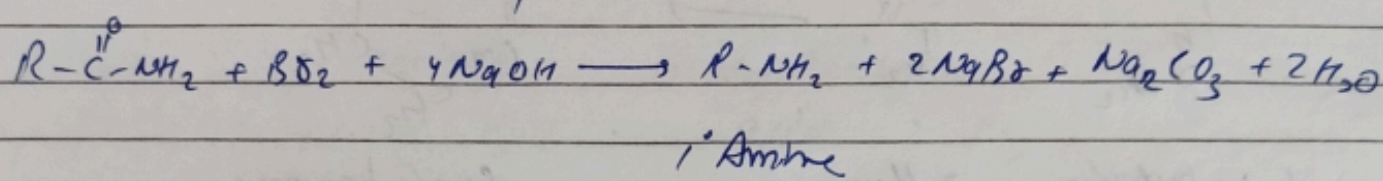
~~Cross Aldol Condensation~~: when aldol condensation
is carried out b/w two different
aldehyde & ketones it is called cross
aldol condensation.

Cannizzaro Rn^1 : we have some aldehyde in which
 α -hydrogen is not present such as
methanal

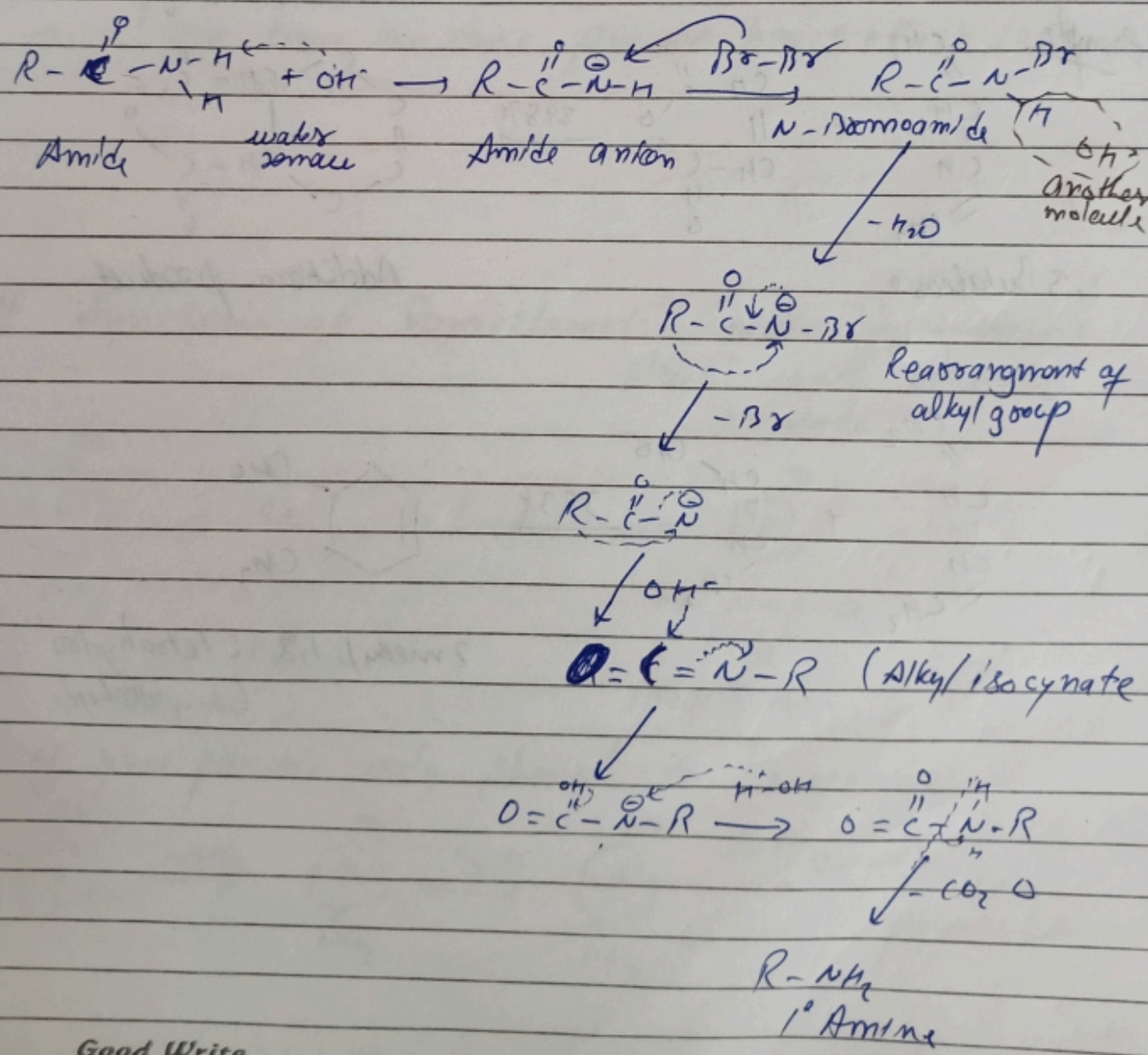


two molecule of Aldehyde one undergo oxidised and forms
Acid
Another molecule Reduced to form Alcohol

Hoffmann's Degradation / Rearrangement: Primary amines can be prepared by treatment of amide with Br_2 and KOH or NaOH . The amine formed contains one carbon atom less than the parent amide.



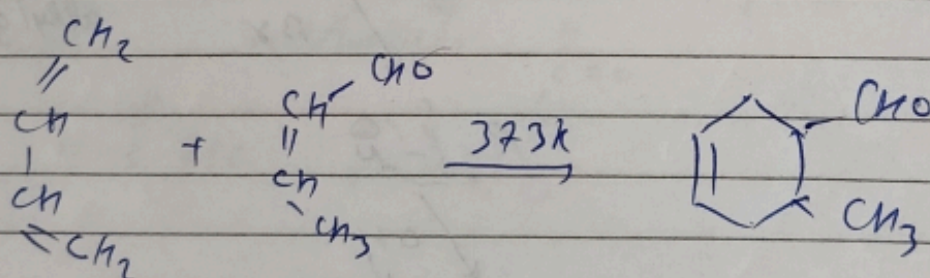
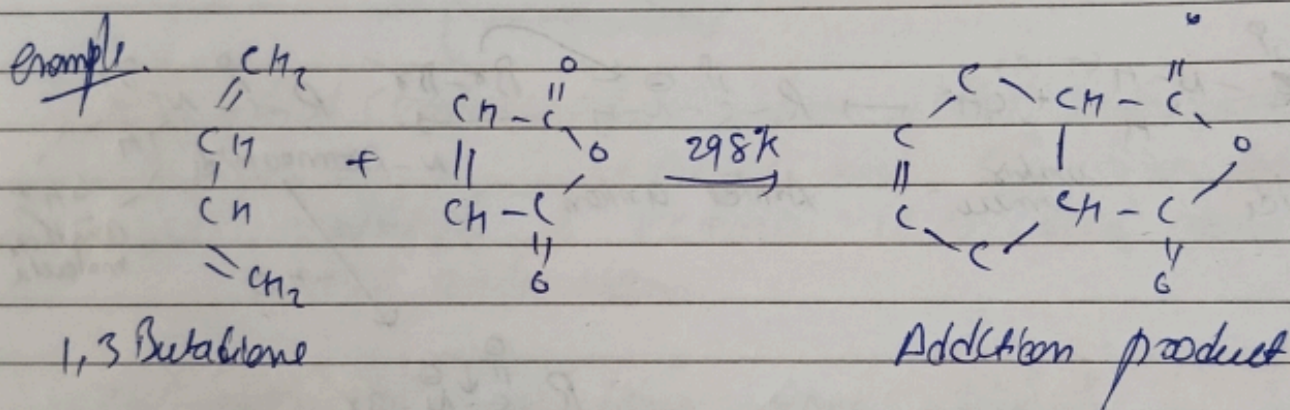
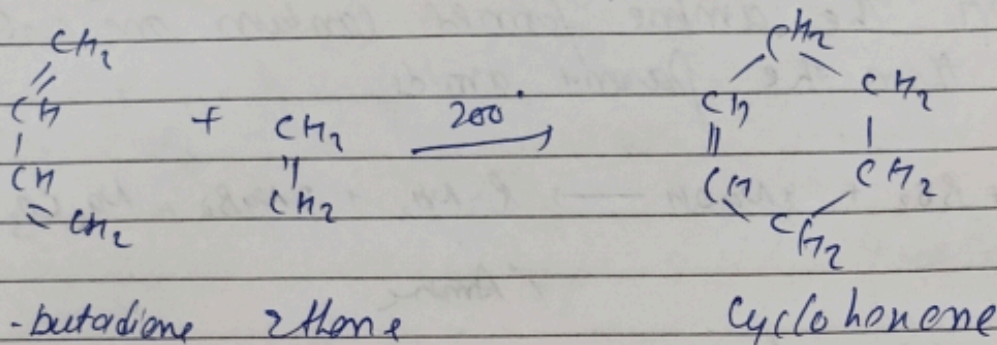
Mechanism. It involves following steps.



cyclization

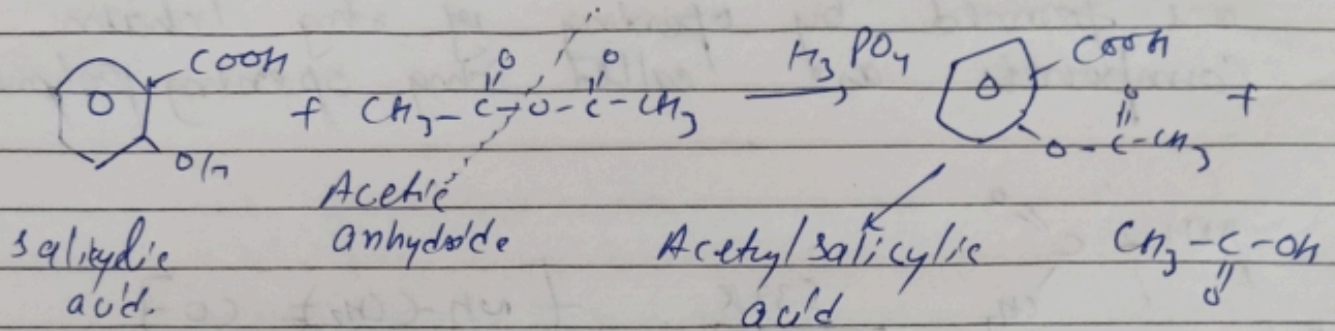
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Diels Alder Reaction (cyclo addition reaction)
1,4 addition of unsaturated comp.
across a conjugated diene.



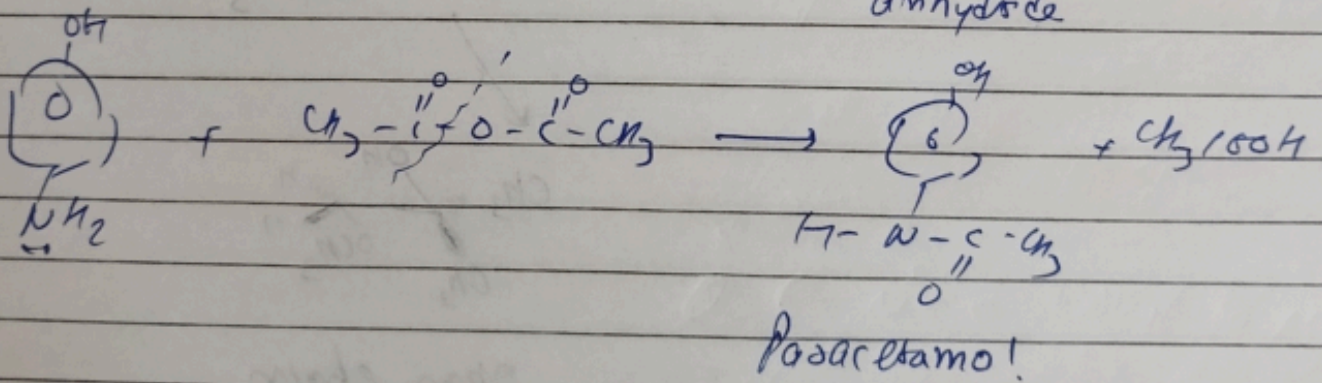
2-methyl, 1,2,3,6-tetrahydro
benzaldehyde

Aspirin synthesis:

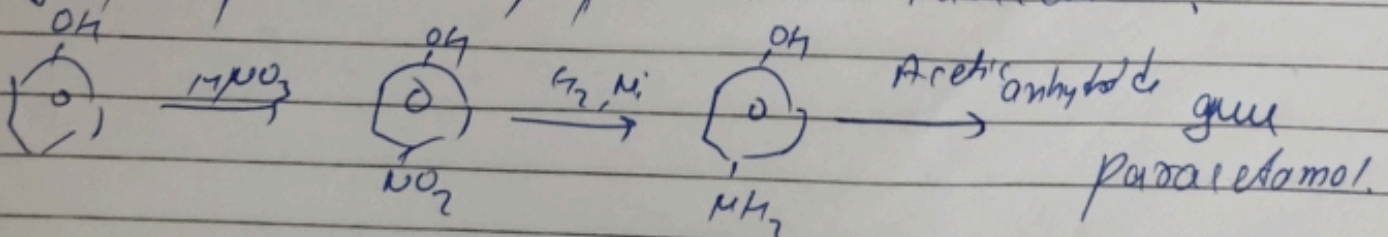


- It is an Antipyretic drug (temp down)
- pain killer
- long term low dose prevent heart attack as a Blood thinner.

Synthesis of Paracetamol: Reaction of 4-amino phenol with acetic anhydride



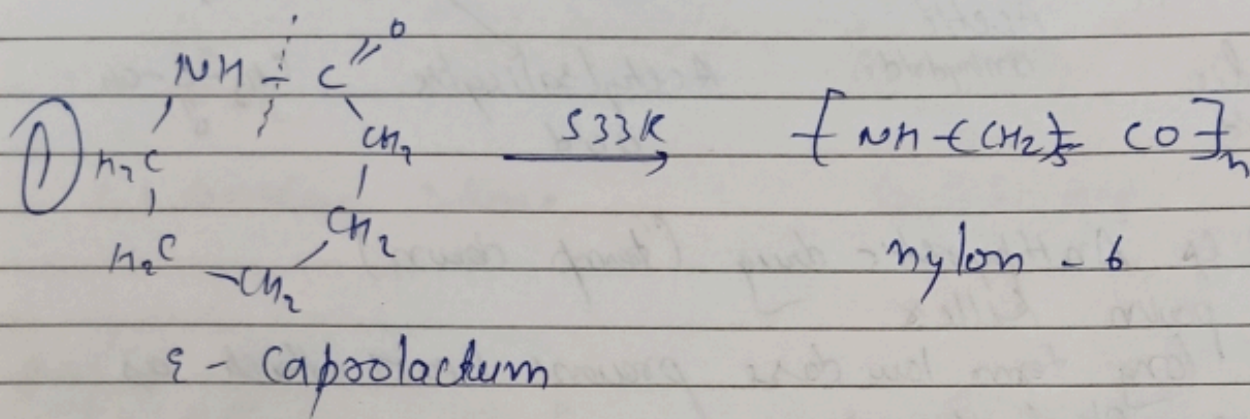
if you provide only phenol, to Paracetamol,



Solvolysis $\rightarrow$ when reactant act as nucleophile is known as Solvolysis.

DATE: ___/___/___
PAGE: _____

Ring Opening Polymerization: The process in which polymers are formed by opening of ring chain compounds are called ring opening polymer.



② Basic ring opening of epoxide:

